

Quiz: Factor Models

1 Index model regression results

The excess return index model has been estimated for Stocks A and B:

	Stock A	Stock B
Regression	$R_A = 1\% + 1.2R_M$	$R_B = -2\% + 0.8R_M$
R^2	0.576	0.436
Residual standard deviation	10.3%	9.1%

- a) Which stock has more firm-specific risk?
- b) Which stock has greater market risk?
- c) For which stock does market movement explain a greater fraction of return variability?
- d) If r_f is constant at 6% and the regression had been run using total returns rather than excess returns, what would the regression intercept for Stock A have been?

Solution:

(a) Firm-specific risk. This is the residual standard deviation, the part of return variability the market factor leaves unexplained: 10.3% for Stock A against 9.1% for Stock B. **Stock A has more firm-specific risk.**

(b) Market risk. Beta measures a security's sensitivity to market-wide uncertainty. Stock A has the higher beta (1.2 against 0.8), so **Stock A carries the greater market risk.**

(c) Explained variability. R^2 is the proportion of total risk that is systematic. Stock A's R^2 of 0.576 exceeds Stock B's 0.436, so **market movements explain a greater fraction of Stock A's return variability.**

(d) Intercept in total returns. Starting from the excess-return specification with $r_f = 6\%$:

$$\begin{aligned} r_A - 6\% &= 1\% + 1.2(r_M - 6\%) \\ r_A &= [1\% + (1 - 1.2) \times 6\%] + 1.2r_M = -0.2\% + 1.2r_M \end{aligned}$$

Regressing total return on total market return would therefore have given an intercept of **-0.2%**.

2 Adjusting and forecasting beta

A stock currently has a beta estimate of 1.24.

- What is the adjusted beta for this stock?
- If you estimate the regression equation $\beta_t = 0.3 + 0.7 \beta_{t-1}$, which describes the evolution of beta over time, what would be your predicted beta for the next year?

Solution:

(a) Adjusted beta. The standard adjustment shrinks the estimate two-thirds of the way from the sample estimate towards the market beta of 1.0:

$$\beta_{\text{adjusted}} = \frac{2}{3} \times 1.24 + \frac{1}{3} \times 1.0 = 1.16$$

(b) Predicted beta.

$$\beta_{\text{predicted}} = 0.3 + 0.7 \times 1.24 = 1.17$$

3 Arbitrage portfolios and the APT

A portfolio manager wishes to use the Arbitrage Pricing Theory (APT) for making investment decisions. The manager evaluates four stocks ($i = 1, 2, 3, 4$) and assumes that two fundamental factors, F_1 and F_2 , generate the return structure. The table below summarises the manager's estimate of the expected return $E(r_i)$, the factor sensitivities β_{i1} and β_{i2} , and the manager's current portfolio weights w_i^0 .

You learn that the set of possible arbitrage weights is $w_1 = -0.280$, $w_2 = 0.043$, $w_3 = -0.065$ and $w_4 = 0.302$.

Stock i	$E(r_i)$	β_{i1}	β_{i2}	w_i^0
1	-0.05	1.5	-1.0	0.32
2	0.26	2.0	2.0	0.48
3	0.166	-0.5	1.0	0.12
4	0.118	1.0	-1.0	0.08

- What is the relationship in general between the portfolio weights w_i^0 and the arbitrage weights w_i ?
- Calculate the new portfolio weights w_i^1 .
- Does this situation give rise to a profitable arbitrage portfolio?

The research department informs you that a possible equilibrium in the stock market is given by the following Arbitrage Pricing Model (APM):

$$E(r_i) = \lambda_0 + \lambda_1 \cdot \beta_{i1} + \lambda_2 \cdot \beta_{i2}$$

where $\lambda_0 = 0.04$, $\lambda_1 = -0.02$ and $\lambda_2 = 0.06$, and β_{i1} and β_{i2} indicate stock i 's sensitivity to Factor 1 and Factor 2 respectively.

- d) What is the interpretation of λ_0 , λ_1 and λ_2 in this model?
- e) Is the manager's expected return on the four stocks (as given in the table) consistent with this equilibrium?
- f) What do you think will happen to future stock prices in this market if the manager's expectations are correct?

Solution:

(a) Portfolio weights and arbitrage weights. If an investor holds a portfolio (with weights w_i^0) whose return is lower than that of another portfolio (with weights w_i^1) carrying the *same* factor exposure, an arbitrage opportunity exists. The investor exploits it by constructing an arbitrage portfolio with weights w_i , so that

$$w_i^1 = w_i^0 + w_i \quad \Longleftrightarrow \quad w_i^0 = w_i^1 - w_i$$

(b) New portfolio weights. Applying $w_i^1 = w_i^0 + w_i$:

$$w_1^1 = 0.32 + (-0.280) = 0.040$$

$$w_2^1 = 0.48 + 0.043 = 0.523$$

$$w_3^1 = 0.12 + (-0.065) = 0.055$$

$$w_4^1 = 0.08 + 0.302 = 0.382$$

(c) Is the arbitrage profitable? Arbitrage pays off if the expected return of the arbitrage portfolio is positive:

$$\begin{aligned} E(r_p^A) &= \sum_{i=1}^4 w_i \cdot E(r_i) \\ &= (-0.280)(-0.05) + (0.043)(0.26) + (-0.065)(0.166) + (0.302)(0.118) \\ &= 0.014 + 0.011 - 0.011 + 0.036 = \mathbf{0.05} \end{aligned}$$

Since $E(r_p^A) > 0$, the arbitrage portfolio is profitable.

(d) Interpretation of the parameters. Within this multi-factor Arbitrage Pricing Model:

- λ_0 is the risk-free return;
- λ_1 is the risk premium compensating investors for exposure to Factor 1 (F_1);
- λ_2 is the risk premium compensating investors for exposure to Factor 2 (F_2).

(e) Consistency with the equilibrium. Under the APM the equilibrium expected return of each stock is:

$$E(r_1) = 0.04 + 1.5(-0.02) - 1.0(0.06) = -0.05$$

$$E(r_2) = 0.04 + 2.0(-0.02) + 2.0(0.06) = 0.12$$

$$E(r_3) = 0.04 - 0.5(-0.02) + 1.0(0.06) = 0.11$$

$$E(r_4) = 0.04 + 1.0(-0.02) - 1.0(0.06) = -0.04$$

Except for Stock 1, the manager's expectations differ from the equilibrium model.

(f) Effect on future prices. Each stock's alpha is the manager's expected return less the equilibrium return:

- Stock 1: $\alpha = -0.05 - (-0.05) = 0$. A zero alpha means the stock is fairly priced, so its price will likely remain unchanged.
- Stock 2: $\alpha = 0.26 - 0.12 = 0.14$. A positive alpha means the stock is under-priced, so its price will likely rise.
- Stock 3: $\alpha = 0.166 - 0.11 = 0.056$. Under-priced, so its price will likely rise.
- Stock 4: $\alpha = 0.118 - (-0.04) = 0.158$. Under-priced, so its price will likely rise.

4 Decomposing total risk

Over the coming period the price of the Tonga stock will either increase by 55% (state 1), fall by 13% (state 2), or remain unchanged (state 3). All three states are equally likely. The market portfolio will either increase by 40% (state 1), remain unchanged (state 2), or fall by 10% (state 3).

State	Probability	Tonga	Market portfolio
1	1/3	+55%	+40%
2	1/3	-13%	0%
3	1/3	0%	-10%

The covariance between the return on Tonga and the market portfolio is 593.33 (computed from the above percentages). The risk-free interest rate is 3%. No dividends are paid.

- a) Using variance as the risk measure, decompose Tonga's total risk into its systematic and unsystematic parts.

Solution:

Working with end-of-period values relative to a base of 100%, the expected value of Tonga is

$$\frac{1}{3}(155\%) + \frac{1}{3}(87\%) + \frac{1}{3}(100\%) = 114\%$$

and its variance is

$$\frac{1}{3}(155\% - 114\%)^2 + \frac{1}{3}(87\% - 114\%)^2 + \frac{1}{3}(100\% - 114\%)^2 = 868.67$$

For the market portfolio,

$$\frac{1}{3}(140\%) + \frac{1}{3}(100\%) + \frac{1}{3}(90\%) = 110\%$$

$$\sigma_M^2 = \frac{1}{3}(140\% - 110\%)^2 + \frac{1}{3}(100\% - 110\%)^2 + \frac{1}{3}(90\% - 110\%)^2 = 466.67$$

Tonga's beta follows from the covariance:

$$\beta = \frac{\sigma_{\text{Tonga},M}}{\sigma_M^2} = \frac{593.33}{466.67} = 1.27$$

The systematic and firm-specific parts of the variance are therefore

$$\text{Systematic: } \beta^2 \times \sigma_M^2 = 1.27^2 \times 466.67 = 754.37$$

$$\text{Firm-specific: } \sigma_e^2 = \sigma_{\text{Tonga}}^2 - \beta^2 \sigma_M^2 = 868.67 - 754.37 = 114.29$$

5 Alpha, risk decomposition and the joint-hypothesis problem

The investment fund Omega holds an equity portfolio with a strong tilt towards value stocks, i.e. stocks with low prices relative to book values. The strategy is to buy the 50 stocks with the lowest price-to-book ratios every year and hold them for one year. Over the last 15 years this strategy has, on average, delivered a statistically significant abnormal return of 5% (α) in excess of the risk-free rate. The CAPM beta of the portfolio over the same period was 0.6. The realised risk premium on the market portfolio was 5% and its variance was 0.04. The R^2 of a regression of the portfolio return on the market return is 0.70.

- Write down the return on the portfolio in terms of a regression equation, using the information given above.
- Decompose the variance of the portfolio into a systematic and an unsystematic component.

Assume now that the expected return is given by the two-factor model

$$E(r_i) - r_f = \beta_i \cdot [E(r_m) - r_f] + h \cdot E(r_{HML})$$

where the expected return on the HML portfolio (r_{HML}) is 5%. Assume that the covariance between the return on the HML portfolio and the market portfolio is zero, and that Omega's factor loading on the HML portfolio (h) is 1.0.

- Write down the return on the portfolio in terms of a regression equation and compare it with the expression in part (a), in terms of the empirical challenge of testing the Efficient Market Hypothesis (EMH).

Solution:

(a) Single-index regression. The excess return of the portfolio is

$$E(r_p - r_f) = \alpha + \beta (r_m - r_f)$$

$$8\% = 5\% + 0.6 (5\%)$$

(b) Variance decomposition. Total risk follows from the definition of beta and the fact that $R^2 = \rho_{P,M}^2$:

$$\beta = \rho_{P,M} \frac{\sigma_P}{\sigma_M} \implies \beta^2 = \rho_{P,M}^2 \frac{\sigma_P^2}{\sigma_M^2} = R^2 \frac{\sigma_P^2}{\sigma_M^2}$$

$$0.6^2 = 0.70 \times \frac{\sigma_P^2}{0.04} \implies \sigma_P^2 = 0.0206$$

The systematic component is

$$\beta^2 \times \sigma_M^2 = 0.6^2 \times 0.04 = 0.0144$$

and the unsystematic component is the remainder:

$$\sigma_e^2 = \sigma_P^2 - \beta^2 \sigma_M^2 = 0.0206 - 0.0144 = 0.0062$$

(c) Two-factor regression and the joint-hypothesis problem.

$$E(r_i) - r_f = \beta_i \cdot [E(r_m) - r_f] + h \cdot E(r_{HML})$$

$$8\% = 0.6 (5\%) + 1.0 (5\%)$$

Note that the two-factor model leaves *no* abnormal return to explain: the same 8% that looked like a 5% alpha under the CAPM is fully accounted for by the HML loading.

If we assume that equilibrium in the market is described by the CAPM, the portfolio is incorrectly priced — as evidenced by the alpha — and the market is inefficient. If instead we assume that equilibrium is described by the two-factor model, the portfolio is correctly priced and the market is efficient. This is the **joint-hypothesis problem** in testing the EMH: we simultaneously test the correctness of the equilibrium model *and* market efficiency, and cannot tell which of the two a rejection refers to.

6 Actual and risk-adjusted probabilities

In the stock market, the market portfolio has an index value of 1,000. Over the next year the index will either increase by 25% or decrease by 15%. The risk-adjusted (fictive) probability of an increase is 0.475. The risk-free interest rate for the period is 4%. A share in the Alfa equity fund is listed at a value of 250. In one year the fund value will be 344 if the market experiences an upturn, or 184 if it experiences a downturn. The risk premium on this equity fund is 8%.

- Based on this information, find the actual probabilities of an increase and a decrease over the next year.
- Based on the Capital Asset Pricing Model (CAPM), what beta does the equity fund have?

- c) Check, for the market portfolio, that discounting future value with actual probabilities and with fictive probabilities yields the same market price today.

Solution:

(a) Actual probabilities. The required return on the Alfa fund is its risk premium plus the risk-free rate: $8\% + 4\% = 12\%$. Writing p^A for the actual probability of an increase and discounting the expected payoff at 12%:

$$250 = \frac{p^A(344) + (1 - p^A)(184)}{1.12}$$

$$280 = 184 + 160 p^A \quad \Rightarrow \quad 96 = 160 p^A \quad \Rightarrow \quad p^A = 0.6$$

The actual probability of an increase is **60%** and of a decrease **40%**.

(b) Beta of the fund. The expected return on the market under the actual probabilities is

$$E(r_M) = 0.6 (25\%) + 0.4 (-15\%) = 9\%$$

Applying the CAPM to the fund's 12% required return:

$$12\% = 4\% + \beta (9\% - 4\%) \quad \Rightarrow \quad \beta = 1.6$$

(c) Two routes to the same price. With actual probabilities the expected payoff is discounted at the market's expected return of 9%:

$$\frac{0.6 (1250) + 0.4 (850)}{1.09} = \frac{1090}{1.09} = 1000$$

With risk-adjusted probabilities the expected payoff is discounted at the risk-free rate of 4%:

$$\frac{0.475 (1250) + 0.525 (850)}{1.04} = \frac{1040}{1.04} = 1000$$

Both give a price of 1,000 today: the risk adjustment can be made either in the discount rate or in the probabilities, but not in both.

7 Standard deviation, covariance and correlation from the index model

Suppose that the index models for Stocks A and B are estimated from excess returns with the following results:

$$\begin{aligned} R_A &= 3\% + 0.7R_M + e_A \\ R_B &= -2\% + 1.2R_M + e_B \\ \sigma_M &= 20\%; \quad R_A^2 = 0.20; \quad R_B^2 = 0.12 \end{aligned}$$

- What is the standard deviation of each stock?
- What is the covariance between each stock and the market index?
- What are the covariance and the correlation coefficient between the two stocks?

Solution:

(a) Standard deviations. The standard deviation of a security follows from

$$\beta_i = \rho_{i,M} \times \frac{\sigma_i}{\sigma_M}, \quad \text{where } \rho_{i,M} = \sqrt{R_i^2}$$

$$0.7 = \sqrt{0.20} \times \frac{\sigma_A}{0.2} \implies \sigma_A = 0.31$$

$$1.2 = \sqrt{0.12} \times \frac{\sigma_B}{0.2} \implies \sigma_B = 0.69$$

(b) Covariance with the market. From $\beta_i = \frac{\text{Cov}_{i,M}}{\sigma_M^2}$:

$$0.7 = \frac{\text{Cov}_{A,M}}{0.2^2} \implies \text{Cov}_{A,M} = 0.028$$

$$1.2 = \frac{\text{Cov}_{B,M}}{0.2^2} \implies \text{Cov}_{B,M} = 0.048$$

(c) Covariance and correlation between the stocks. In the index model the only source of comovement between two stocks is the market factor, so

$$\text{Cov}_{A,B} = \beta_A \times \beta_B \times \sigma_M^2 = 0.7 \times 1.2 \times 0.2^2 = 0.0336$$

The correlation is then

$$\text{Corr}_{A,B} = \rho_{A,M} \times \rho_{B,M} = \sqrt{0.20} \times \sqrt{0.12} = 0.15$$

or equivalently

$$\text{Corr}_{A,B} = \frac{\beta_A \times \beta_B \times \sigma_M^2}{\sigma_A \times \sigma_B} = \frac{0.7 \times 1.2 \times 0.2^2}{0.31 \times 0.69} = 0.15$$

8 Applying the CAPM

The risk-free rate of interest available in the market is 6% and the expected excess rate of return of the market index is 16%.

- A share of stock sells for USD 50 today. It will pay a dividend of USD 6 per share at the end of the year. Its beta is 1.2. What do investors expect the stock to sell for at the end of the year?

- b) A stock has an expected rate of return of 4%. What is its beta?
- c) You are evaluating a firm with an expected perpetual cash flow of USD 1,000, but you are unsure of its risk. If you think the beta of the firm is 0.5 when it is actually 1.0, how much more will you offer for the firm compared with its true worth?

Solution:

(a) Expected price from the CAPM. The expected return required by the CAPM is

$$E(r) = 6\% + 1.2 \times 16\% = 25.2\%$$

The expected end-of-year price follows from the definition of the holding period return:

$$r_{\text{stock}} = \frac{P_1 - P_0 + \text{Dividend}}{P_0}$$

$$0.252 = \frac{P_1 - 50 + 6}{50} \quad \Rightarrow \quad P_1 - 44 = 12.6 \quad \Rightarrow \quad P_1 = 56.6$$

(b) Beta from an expected return.

$$6\% + \beta \times 16\% = 4\% \quad \Rightarrow \quad \beta = -0.13$$

A stock with a negative beta is negatively correlated with the market: it tends to rise when the market falls and vice versa. Such assets may be considered “safe-haven assets”.

(c) Mispricing a firm by misjudging its beta. The value of the firm is its perpetual cash flow discounted at the required rate of return, $V_0 = \frac{CF_{\infty}}{r}$.

If $\beta = 0.5$:

$$r = 6\% + 0.5 \times 16\% = 14\% \quad \Rightarrow \quad V_0 = \frac{1000}{0.14} = \text{USD } 7,143$$

If $\beta = 1.0$:

$$r = 6\% + 1.0 \times 16\% = 22\% \quad \Rightarrow \quad V_0 = \frac{1000}{0.22} = \text{USD } 4,545$$

By underestimating the beta you underestimate the required return, discount too gently, and pay

$$7,143 - 4,545 = \text{USD } 2,598$$

more than the firm is worth.

9 The CAPM and the security market line

Suppose that the rate of return on short-term government securities (perceived to be risk free) is about 5%. Suppose also that the expected rate of return required by the market for a portfolio with a beta of 1 is 12%. According to the Capital Asset Pricing Model:

- a) What is the expected rate of return on the market portfolio?
- b) What would be the expected rate of return on a stock with $\beta = 0$?
- c) Suppose that you consider buying a share of stock at USD 40. The stock is expected to pay USD 3 in dividends next year and you expect it to sell then for USD 41. The stock's risk has been evaluated at $\beta = -1.5$. Is the stock overpriced or underpriced?

Solution:

(a) **The market portfolio.** A stock with a beta equal to 1 has the same expected return as the market portfolio, so the expected return on the market portfolio is **12%**. Verify:

$$12\% = 5\% + 1 \times (12\% - 5\%)$$

(b) **A zero-beta stock.** A stock with a beta of zero carries no systematic risk and therefore earns the risk-free rate, **5%**. Verify:

$$5\% = 5\% + 0 \times (12\% - 5\%)$$

(c) **Overpriced or underpriced?** The return required by the CAPM is

$$E(r) = 5\% + (-1.5) \times (12\% - 5\%) = -5.5\%$$

while the return the investor expects is

$$\frac{41 - 40 + 3}{40} = 10\%$$

The gap between the two is a positive alpha of

$$10\% - (-5.5\%) = 15.5\%$$

which means the stock is **underpriced**. Its negative beta makes it valuable as a hedge, so the market requires a return *below* the risk-free rate to hold it — yet it is expected to deliver 10%.

10 A two-factor economy

Assume that there are two independent economic factors, F_1 and F_2 . The risk-free rate is 6% and all stocks have independent firm-specific components with a standard deviation of 45%. Portfolios A and B are both well diversified, with the properties below. What is the expected return–beta relationship in this economy?

Portfolio	Beta on F_1	Beta on F_2	Expected return
A	1.5	2.0	31%
B	2.2	−0.2	27%

Solution:

Because both portfolios are well diversified, their firm-specific risk is negligible and expected returns are governed by the multi-factor model

$$E(r_i) = r_f + \beta_{i1}F_1 + \beta_{i2}F_2$$

Substituting each portfolio gives two equations in the two factor risk premia:

$$\text{Portfolio A: } 31\% = 6\% + 1.5 F_1 + 2.0 F_2$$

$$\text{Portfolio B: } 27\% = 6\% + 2.2 F_1 - 0.2 F_2$$

Solving the system gives $F_1 = 10\%$ and $F_2 = 5\%$, so the expected return–beta relationship is

$$E(r_i) = 6\% + 10\% \beta_{i1} + 5\% \beta_{i2}$$

11 The risk-free rate in a one-factor economy

Assume that portfolios A and B are well diversified, with $E(r_A) = 12\%$ and $E(r_B) = 9\%$. If the economy has only one factor, and $\beta_A = 1.2$ and $\beta_B = 0.8$, what must the risk-free rate be?

Solution:

Since portfolios A and B are well diversified, their expected returns are given by $E(r_i) = r_f + \beta_i F$:

$$E(r_A) : 12\% = r_f + 1.2 F$$

$$E(r_B) : 9\% = r_f + 0.8 F$$

The factor premium F is unknown, so we solve the two equations simultaneously. Subtracting the second from the first eliminates r_f :

$$12\% - 9\% = (1.2 - 0.8) F \quad \implies \quad 0.4 F = 3\% \quad \implies \quad F = 7.5\%$$

Substituting back into the equation for $E(r_A)$:

$$12\% = r_f + 1.2 \times 7.5\% \quad \Rightarrow \quad r_f = 3\%$$

Verify with portfolio B: $9\% = 3\% + 0.8 \times 7.5\%$, as given.

12 A multi-factor model with macroeconomic risk premia

You are given the following information about the factors in the multi-factor model of a particular stock: the inflation risk premium (INF) is 6% with a factor loading of 1.2; the industrial production risk premium (IND) is 8% with a factor loading of 0.5; and the oil price risk premium (OIL) is 3% with a factor loading of 0.3. Treasury bills currently offer a 6% yield.

Find the expected rate of return on this stock if the market views it as correctly priced.

Solution:

The equilibrium model of the stock's return is

$$E(r_i) = r_f + \beta_{i,INF} INF + \beta_{i,IND} IND + \beta_{i,OIL} OIL$$

$$E(r_i) = 6\% + 1.2 (6\%) + 0.5 (8\%) + 0.3 (3\%) = 18.1\%$$

13 Arbitrage in a one-factor economy

Consider the following data for a one-factor economy. Both portfolios are well diversified.

Portfolio	$E(r)$	Beta
A	12%	1.2
F	6%	0.0

Suppose that another portfolio E exists, which is well diversified with a beta of 0.6 and an expected return of 8%. Would an arbitrage opportunity exist? If so, what would the arbitrage strategy be?

Solution:

Since portfolios A and F are well diversified, their expected returns satisfy $E(r_i) = r_f + \beta_i F$. Portfolio F has a beta of 0, so its return is the risk-free rate, $r_f = 6\%$. The factor premium then follows from portfolio A:

$$12\% = 6\% + 1.2 F \quad \Rightarrow \quad F = 5\%$$

Under this equilibrium, a well-diversified portfolio with a beta of 0.6 should have an expected return of

$$E(r) = 6\% + 0.6 (5\%) = 9\%$$

Portfolio E offers only 8%, so it is incorrectly priced and **an arbitrage opportunity exists**.

The strategy is to build a portfolio with the same beta of 0.6 out of the correctly priced portfolios A and F. Taking 50% in A and 50% in F gives a new portfolio N with

$$\beta_N = 0.5 (1.2) + 0.5 (0) = 0.6$$

$$E(r_N) = 0.5 (12\%) + 0.5 (6\%) = 9\%$$

Portfolio N matches E's factor exposure exactly but earns more. Selling portfolio E short and buying portfolio N with the proceeds therefore yields a risk-free profit of

$$9\% - 8\% = 1\%$$